

संभव BATCH - CRASH COURSE

Answer Key & Solutions: Geography Moderate Module

Expert Guidance: **Vipulendra Tripathi Sir**

Zero Error Verified

Official Step-by-Step Analytical Answer Key for the 50 Moderate Geography MCQs.

Question 1

Correct Answer: Venus

Geographical Explanation:

Venus is often called the 'Morning Star' or 'Evening Star' because it is frequently the brightest celestial object in the sky (other than the moon) visible just before sunrise or just after sunset.

Question 2

Correct Answer: Mars and Jupiter

Geographical Explanation:

The asteroid belt is a torus-shaped region in the Solar System, located roughly between the orbits of the planets Mars and Jupiter, containing numerous irregularly shaped bodies called asteroids.

Question 3

Correct Answer: Ganymede

Geographical Explanation:

Ganymede, which orbits Jupiter, is the largest and most massive moon in the Solar System. It is even larger than the planet Mercury.

Question 4

Correct Answer: Venus

Geographical Explanation:

Unlike most planets in the solar system, Venus and Uranus rotate on their axes from East to West. This is known as retrograde rotation.

Question 5

Correct Answer: Nuclear Fusion

Geographical Explanation:

The Sun's immense energy is generated by nuclear fusion in its core, where hydrogen nuclei fuse together under extreme temperature and pressure to form helium, releasing vast amounts of energy.

Question 6

Correct Answer: Saturn

Geographical Explanation:

Saturn has the lowest average density of any planet in our solar system (about 0.687 g/cm^3), which is less than the density of water (1 g/cm^3).

Question 7

Correct Answer: Corona

Geographical Explanation:

The outermost layer of the Sun's atmosphere is called the Corona. It extends millions of kilometers into space and is most easily seen during a total solar eclipse.

Question 8

Correct Answer: Venus

Geographical Explanation:

Despite Mercury being closer to the Sun, Venus is the hottest planet. Its thick atmosphere, composed primarily of carbon dioxide, traps heat in a runaway greenhouse effect, keeping surface temperatures around 465°C .

Question 9

Correct Answer: The habitable zone around a star where liquid water can exist

Geographical Explanation:

The 'Goldilocks Zone' (or habitable zone) refers to the optimal range of distance from a star where conditions are neither too hot nor too cold, allowing liquid water to exist on a planet's surface.

Question 10

Correct Answer: 76 years

Geographical Explanation:

Halley's Comet is a short-period comet visible from Earth every 75–76 years. It is the only known short-period comet that is regularly visible to the naked eye from Earth.

Question 11

Correct Answer: Mars

Geographical Explanation:

Mars is widely known as the 'Red Planet' because its surface is covered with iron oxide (rust) dust, giving it a distinct reddish appearance.

Question 12

Correct Answer: Mercury

Geographical Explanation:

Mercury and Venus are the only two major planets in our solar system that do not have any natural satellites (moons).

Question 13

Correct Answer: Igneous rocks

Geographical Explanation:

Igneous rocks are formed through the cooling and solidification of magma (below the surface) or lava (on the surface). They are considered primary rocks.

Question 14

Correct Answer: Sedimentary rocks

Geographical Explanation:

Fossils are predominantly found in sedimentary rocks, as the gradual accumulation and compaction of sediments gently bury and preserve the remains of dead organisms without destroying them via extreme heat.

Question 15

Correct Answer: Quartzite

Geographical Explanation:

When sandstone (a clastic sedimentary rock rich in quartz) is subjected to the high heat and pressure of regional or contact metamorphism, it transforms into quartzite.

Question 16

Correct Answer: Granite

Geographical Explanation:

Granite is a classic example of an intrusive (plutonic) igneous rock. It forms when magma cools slowly deep beneath the Earth's surface, allowing large, visible crystals to form.

Question 17

Correct Answer: Sedimentary rocks

Geographical Explanation:

Fossil fuels like coal and petroleum are typically found in sedimentary rocks. They form from the compressed and heated remains of ancient organic matter buried under layers of sediment.

Question 18

Correct Answer: Marble

Geographical Explanation:

Limestone is a sedimentary rock consisting largely of calcium carbonate. Under conditions of intense heat and pressure, it recrystallizes and metamorphoses into marble.

Question 19

Correct Answer: Pumice

Geographical Explanation:

Pumice is an extrusive volcanic rock with a highly vesicular (porous) texture formed from frothy lava. It is so light and porous that it can float on water.

Question 20

Correct Answer: Lithification

Geographical Explanation:

Lithification is the complex geological process involving compaction and cementation, by which loose, unconsolidated sediments are converted into solid sedimentary rock.

Question 21

Correct Answer: Extrusive igneous rock

Geographical Explanation:

Basalt is a fine-grained, dark-colored extrusive igneous rock. It forms from the rapid cooling of low-viscosity lava at or very near the Earth's surface.

Question 22

Correct Answer: Chalk

Geographical Explanation:

Chalk is a soft, white, porous sedimentary rock, a form of limestone composed mostly of the mineral calcite, which derives from the skeletal remains of microscopic marine organisms.

Question 23

Correct Answer: Rock cycle

Geographical Explanation:

The rock cycle is a continuous, dynamic geological concept that describes the time-consuming transitions through geological time among the three main rock types: sedimentary, metamorphic, and igneous.

Question 24

Correct Answer: Slate

Geographical Explanation:

Shale is a fine-grained sedimentary rock. Under low-grade metamorphic conditions (moderate heat and directed pressure), it typically transforms into slate, which exhibits strong foliation.

Question 25

Correct Answer: Pacific Ocean

Geographical Explanation:

The Ring of Fire is a major area in the basin of the Pacific Ocean where many earthquakes and volcanic eruptions occur. It is associated with a nearly continuous series of oceanic trenches and volcanic arcs.

Question 26

Correct Answer: Caldera

Geographical Explanation:

A caldera is a large, basin-like depression created by the collapse of a volcano into itself following a massive, explosive eruption that partially or completely empties its underlying magma chamber.

Question 27

Correct Answer: Barren Island

Geographical Explanation:

Barren Island, located in the Andaman Sea (Andaman and Nicobar Islands), is the only confirmed active volcano in the Indian subcontinent.

Question 28

Correct Answer: Shield volcano

Geographical Explanation:

Shield volcanoes are broad, domed volcanoes with gently sloping sides. They are built almost entirely of highly fluid (low viscosity) basaltic lava flows, such as Mauna Loa in Hawaii.

Question 29

Correct Answer: Dyke

Geographical Explanation:

A dyke is an intrusive igneous rock body that forms when magma is injected into a fracture or fissure and cuts vertically or diagonally across existing rock layers (discordant intrusion).

Question 30

Correct Answer: Water vapour

Geographical Explanation:

Water vapour (H₂O) is the most abundant volcanic gas emitted into the atmosphere during an eruption, often constituting more than 60% of the total gas emissions by volume.

Question 31

Correct Answer: Sill

Geographical Explanation:

A sill is a tabular sheet intrusion of magma that has intruded horizontally between and parallel to older layers of sedimentary rock, beds of volcanic lava, or along the direction of foliation in metamorphic rock.

Question 32

Correct Answer: Composite volcano (Stratovolcano)

Geographical Explanation:

Mount Fuji is a classic stratovolcano (composite volcano), characterized by its steep, symmetrical cone built up by many alternating layers (strata) of hardened lava, pumice, and volcanic ash.

Question 33

Correct Answer: Lava

Geographical Explanation:

Magma is molten rock stored beneath the Earth's surface. Once it erupts and flows onto the Earth's surface through a volcano or fissure, it is termed lava.

Question 34

Correct Answer: Stromboli

Geographical Explanation:

Stromboli, a volcanic island in Italy, is in a state of nearly continuous, mild eruption. Its regular incandescent nighttime explosions have earned it the nickname 'Lighthouse of the Mediterranean'.

Question 35

Correct Answer: Weathering is the in-situ breakdown of rocks; erosion involves the transportation of the material.

Geographical Explanation:

Weathering is the static process of rock disintegration and decomposition in place (in-situ). Erosion is dynamic, involving the picking up and transportation of this weathered material by agents like wind, water, or ice.

Question 36

Correct Answer: Oxidation

Geographical Explanation:

Oxidation is a chemical weathering process where minerals in a rock react with oxygen in the presence of water, commonly leading to the rusting of iron-bearing rocks. Frost wedging and exfoliation are mechanical.

Question 37

Correct Answer: Exfoliation

Geographical Explanation:

Exfoliation (or unloading) is a physical weathering process where the removal of overlying rock reduces pressure, causing the underlying rock to expand and shed outer layers like the skin of an onion.

Question 38

Correct Answer: Hot and Humid

Geographical Explanation:

Chemical weathering involves chemical reactions (like hydration, oxidation, and carbonation) that proceed much more rapidly in the presence of abundant moisture and high temperatures (tropical environments).

Question 39

Correct Answer: Rivers

Geographical Explanation:

A V-shaped valley is a classic youthful stage feature of fluvial (river) erosion. The river's steep gradient causes strong vertical down-cutting, forming the V-shape.

Question 40

Correct Answer: Glaciers

Geographical Explanation:

U-shaped valleys, or glacial troughs, are formed by the process of glaciation. As a massive glacier moves down a valley, its immense weight and abrasive power pluck and scrape the valley floor and sides, widening it into a 'U' shape.

Question 41

Correct Answer: Wind

Geographical Explanation:

Mushroom rocks (Zeugen) are created by Aeolian (wind) erosion. Wind transports sand and abrasive materials close to the ground, eroding the base of a rock mass faster than its upper part.

Question 42

Correct Answer: Delta

Geographical Explanation:

A delta is a prominent depositional landform created at the mouth of a river where it flows into an ocean, sea, or lake. As the river's velocity drops, it drops its sediment load. Gorges, canyons, and waterfalls are erosional.

Question 43

Correct Answer: Groundwater

Geographical Explanation:

Stalactites and stalagmites are speleothems found in limestone caves. They are formed by the slow precipitation of calcium carbonate from mineral-rich groundwater dripping from the cave ceiling.

Question 44

Correct Answer: Mass wasting

Geographical Explanation:

Mass wasting (or mass movement) refers to the downslope movement of rock, regolith, and soil under the direct, dominant influence of gravity. It includes landslides, rockfalls, and mudflows.

Question 45

Correct Answer: An ox-bow lake

Geographical Explanation:

An ox-bow lake is a U-shaped body of water that forms when a wide meander from the main stem of a river is cut off, creating a free-standing body of water.

Question 46

Correct Answer: Moraines

Geographical Explanation:

Moraines are accumulations of dirt, rocks, and unstratified glacial till deposited by a glacier. Cirques, horns, and aretes are all features formed by glacial erosion, not deposition.

Question 47

Correct Answer: Wind

Geographical Explanation:

Loess is a predominantly silt-sized, wind-blown (Aeolian) sediment. Extensive loess deposits are highly fertile and are found in regions like the Loess Plateau in China.

Question 48

Correct Answer: Limestone

Geographical Explanation:

Sinkholes are characteristic features of karst topography. They form predominantly in regions with carbonate bedrock, such as limestone, which is highly susceptible to chemical dissolution by groundwater.

Question 49

Correct Answer: Carbonation

Geographical Explanation:

Carbonation occurs when carbon dioxide dissolves in rainwater to form weak carbonic acid. This acid reacts with the calcium carbonate in limestone, gradually dissolving it.

Question 50

Correct Answer: Lahar

Geographical Explanation:

A lahar is a violent, destructive type of mudflow or debris flow composed of a slurry of pyroclastic material, rocky debris, and water. The material flows rapidly down a volcano, often triggered by melting snow/ice or heavy rain.

End of Answer Key - "संभव Batch" - Prepared for Excellence with Zero Errors.